ContextST: Coordinate-Frame-Invariant Multi-Context Attention for Spatial Gene Expression Prediction from Histology

Anonymous authors
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Abstract

Spatial transcriptomics (ST) maps gene expression within intact tissue but remains substantially less accessible than routine hematoxylin-and-eosin histology. Inferring spatial expression from histology requires a model to connect the morphology of each capture spot with both its tissue neighbourhood and the broader slide context, without becoming sensitive to an arbitrary coordinate frame. We present ContextST, a compact multi-context architecture that predicts gene expression for every spot on a slide in one forward pass. Starting from frozen histology patch embeddings, each ContextST block processes three complementary contexts in parallel: distance-biased attention over spatial k-nearest neighbours captures local tissue organization, full self-attention models slide-wide spot interactions, and attention pooling forms a shared slide representation. The three signals are fused residually before a direct gene-expression regression head. Coordinates enter the network only through radial-basis encodings of pairwise distances; therefore, for fixed image features, the spatial pathway is invariant to translations, rotations, and reflections of the coordinate frame. We evaluate ContextST on HEST-1k using matched features, gene panels, training budgets, and nested validation across intra-cohort, pooled cross-sample, and leave-one-organ-out settings. ContextST achieves a mean intra-cohort Pearson correlation of 0.470, a 4.9% relative improvement over the strongest baseline in the main comparison, and reaches 0.490 under leave-one-organ-out transfer. Factorial ablations identify distance-aware local attention as the primary contributor, while the complete local–global–slide model performs best. With 4.88M parameters and 64.7 ms inference for a 10,000-spot slide on an A100, ContextST provides an efficient, coordinate-frame-invariant approach to spatial gene expression prediction from histology.

1 Introduction

Spatial transcriptomics (ST) measures gene expression together with tissue location, enabling the study of spatial heterogeneity, tissue organization, and cellular interactions Ståhl et al. (2016). Its cost and experimental burden, however, limit the number and diversity of profiled specimens. Hematoxylin-and-eosin (H&E) whole-slide images are routinely available and encode morphology associated with tissue composition and molecular state. Predicting spatial expression from H&E images could therefore extend molecular analysis to samples without ST measurements.

Prior approaches span independent patch regression He et al. (2020), spatial transformers and graphs Pang et al. (2021); Zeng et al. (2022), retrieval through cross-modal alignment Xie et al. (2023), and conditional generation Zhu et al. (2025). These models answer an important question—how accurately can expression be inferred within a studied tissue cohort?—but usually train and test on slides from the same organ. Such evaluation does not isolate transfer across organs, where tissue architecture, cellular composition, staining, and expression distributions all shift.

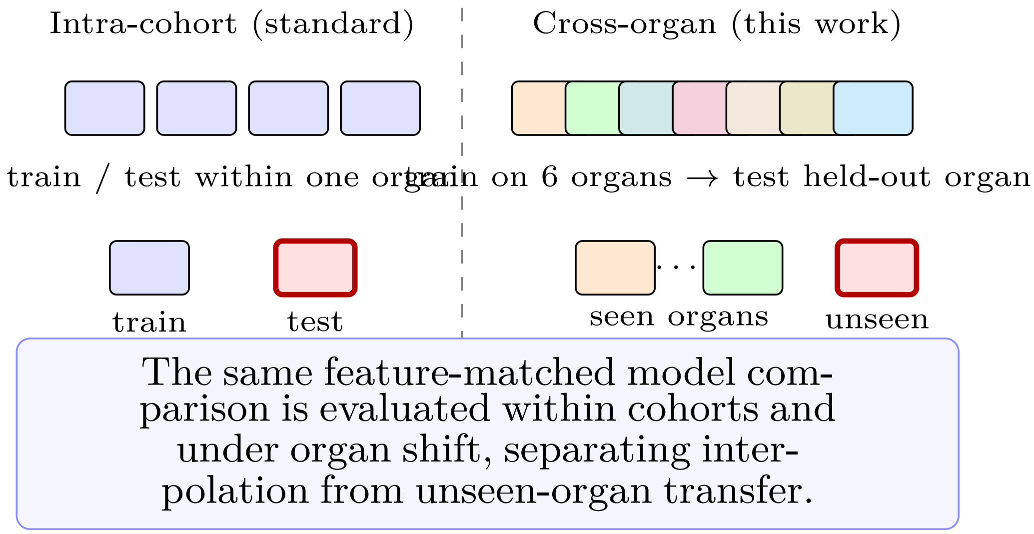

Figure 1: Two evaluation regimes. Standard benchmarks train and test within one tissue cohort (left). We additionally test the harder cross-organ transfer (right), where a model must predict expression for slides from organs unseen during training. ContextST is evaluated in both without per-cohort redesign.

We study a complementary question: how far can a compact point-prediction model generalize when its spatial inductive bias is explicit and its evaluation is controlled? ContextST predicts a complete slide in one pass from frozen pathology features and spot coordinates. Each layer integrates local neighbourhood structure, global spot-to-spot interactions, and a learned slide summary. Coordinates are used only through pairwise Euclidean distances, making the spatial pathway invariant to rigid coordinate transformations and reflections without test-time frame averaging.

We evaluate three distinct regimes (Fig. 1): intra-cohort prediction, pooled cross-validation on unseen samples from represented organs, and leave-one-organ-out (LOOO) transfer to an unseen organ. Under feature-matched nested validation, ContextST obtains the highest mean intra-cohort Pearson correlation and remains competitive in LOOO transfer. These results position direct regression as a strong, efficient point-prediction baseline; they do not replace generative methods when the goal is to model the full conditional distribution of expression.

Our contributions are:

- a local–global spatial transformer whose coordinate pathway is provably invariant to translations, rotations, and reflections;
- a controlled HEST-1k evaluation that distinguishes within-cohort accuracy, generalization to new samples from known organs, and transfer to an unseen organ; and
- a factorial analysis of spatial context together with measured parameter, latency, and memory scaling for single-pass slide-level inference.

2 Related Work

2.1 Histology-based gene expression prediction

ST-Net He et al. (2020) established patch-level regression from histology to expression. HisToGene Pang et al. (2021) and Hist2ST Zeng et al. (2022) subsequently introduced slide-level attention and spatial graphs, while TRIPLEX Chung et al. (2024) combines local, neighbourhood, and global morphology. BLEEP Xie et al. (2023) follows a different

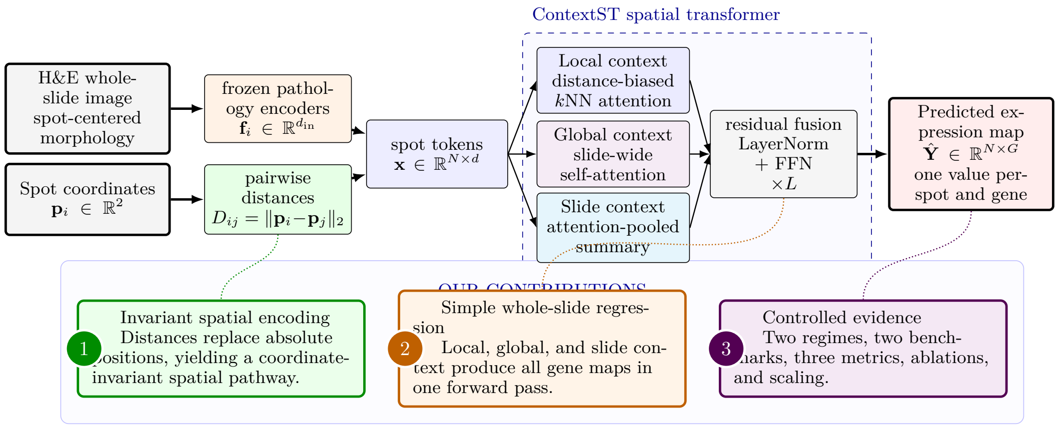

Figure 2: ContextST overview. The model receives frozen pathology morphology features and spot coordinates. Coordinates affect the network only through pairwise distances. Each layer combines local, global, and slide-level context before a linear head predicts the expression of G genes at all N spots in one forward pass.

strategy: it aligns image and expression embeddings and retrieves expression from similar training examples. More recent models incorporate higher-order cellular interactions Sun et al. (2026), dual-path discrimination Song et al. (2026), or hyperbolic cross-modal structure Zhang et al. (2026). Because their original evaluations vary in features, gene panels, splits, and model-selection procedures, published scores do not by themselves isolate architectural gains.

STEM Zhu et al. (2025) frames expression inference as conditional generation rather than deterministic regression. This distinction is important: a generator can represent multiple expression profiles compatible with similar morphology, whereas ContextST estimates one profile per spot. Our objective is therefore narrower. We ask whether an inexpensive point predictor can provide a strong accuracy and transfer baseline under a matched protocol; we do not claim to model biological uncertainty or the full conditional expression distribution.

2.2 Geometric structure in spatial models

Spatial prediction depends on the coordinate representation supplied to a model. Absolute position embeddings can encode location within a particular coordinate frame, but their output may change when an equivalent tissue layout is translated, rotated, or reflected. Geometric networks address this issue by constructing interactions from relative quantities Satorras et al. (2021). ContextST uses an invariant construction in which coordinates enter its spatial pathway only through pairwise Euclidean distances. Because these distances are preserved by translations, rotations, and reflections, no transformed coordinate frames or prediction averaging are required. This guarantee applies to the spatial pathway for fixed histology features; it does not imply that the frozen image encoder is invariant to rotations of the image itself.

2.3 Generalization across samples and organs

HEST-1k Jaume et al. (2024) spans many cohorts and organs, yet its standard predictive benchmark is organized within cohorts. A model can therefore exploit organ-specific morphology, expression statistics, or acquisition effects. Cross-sample benchmarking also identifies distribution shift as a central challenge others (2025). We separate two forms of multi-organ evaluation: pooled cross-validation holds out samples while retaining their organ types in training, whereas LOOO withholds an entire organ. The latter is a stricter test of anatomical transfer.

3 Method

3.1 Problem setup

For a slide with N spots, let $\mathbf{F} = [\mathbf{f}_1, \ldots, \mathbf{f}_N]^\top \in \mathbb{R}^{N \times d_{\text{in}}}$ denote frozen pathology features and $\mathbf{P} = [\mathbf{p}_1, \ldots, \mathbf{p}_N]^\top \in \mathbb{R}^{N \times 2}$ the corresponding spot coordinates. The target $\mathbf{Y} \in \mathbb{R}^{N \times G}$ contains log-normalized expression for G genes. We learn $f_\theta : (\mathbf{F}, \mathbf{P}) \mapsto \hat{\mathbf{Y}}$ and process each slide independently; the model therefore supports a variable number of spots and requires no expression measurements at test time. In the feature-matched main comparisons, $\mathbf{f}_i$ concatenates frozen UNI Chen et al. (2024) and CONCH Lu et al. (2024) embeddings ($d_{\text{in}} = 1536$).

3.2 Coordinate-invariant spatial encoding

We first compute the pairwise Euclidean distance matrix $D_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2$. For each spot i, we retain the indices $\mathcal{N}(i)$ and distances of its k nearest neighbours, excluding the spot itself. Distances are expanded into M radial-basis features,

$$\boldsymbol{\phi}(D_{ij}) = \left[\exp\left(-\gamma_m D_{ij}^2\right)\right]_{m=1}^{M}, \tag{1}$$

where the fixed bandwidth parameters γ_m are logarithmically spaced. These features influence attention weights but are never added to the spot embeddings as absolute positions.

This construction yields coordinate-system invariance. For any orthogonal matrix R and translation $\mathbf{t}$,

$$\|R\mathbf{p}_i + \mathbf{t} - (R\mathbf{p}_j + \mathbf{t})\|_2 = \|\mathbf{p}_i - \mathbf{p}_j\|_2. \tag{2}$$

The neighbour graph, radial-basis features, and predictions are therefore unchanged by translations, rotations, and reflections of the coordinates, up to floating-point error, when the image features are held fixed. The result concerns transformations of the coordinate system and does not assert rotation invariance of the frozen histology encoder.

3.3 The ContextST layer

The input features are projected to hidden width d, $\mathbf{X}^{(0)} = \mathbf{F}W_{\text{in}} + \mathbf{b}_{\text{in}}$. The same neighbour graph and radial-basis features are reused in all L transformer layers. At layer ℓ, a pre-normalized representation $\mathbf{H} = \mathrm{LN}(\mathbf{X}^{(\ell)})$ is passed through three context pathways.

Local distance-biased attention. Each spot attends to its k neighbours with a learned, distance-dependent bias for each attention head. For head h, the attention weights are

$$\alpha_{ij}^h = \mathrm{softmax}_{j \in \mathcal{N}(i)} \left(\frac{(\mathbf{q}_i^h)^\top \mathbf{k}_j^h}{\sqrt{d_h}} + b_h(\boldsymbol{\phi}(D_{ij})) \right), \tag{3}$$

where $d_h = d/H$ for H heads. The local output is the concatenation of the weighted value sums, followed by a learned output projection. Morphological similarity is captured by the query–key term, while spatial distance contributes through the learned radial-basis bias.

Global self-attention. In parallel, standard multi-head self-attention Vaswani et al. (2017) allows every spot to attend to all N spots. This pathway captures long-range dependencies that may not be connected in the local graph. It uses morphology features only and is permutation-equivariant with respect to spot order.

Attention-pooled slide context. A learned scalar scoring function produces $a_i = \mathrm{softmax}_i(s(\mathbf{h}_i))$. The slide representation is

$$\mathbf{g} = W_g \left(\sum_{i=1}^{N} a_i \mathbf{h}_i \right) + \mathbf{b}_g, \tag{4}$$

and is broadcast to every spot. Unlike a learned class token, this representation is computed directly from the slide's spot features and summarizes its morphological composition.

Table 1: Intra-cohort gene expression prediction on HEST-1k, per cohort (Pearson ↑). Mean_{std} over three seeds with nested-validation; best per row in bold. All methods use identical frozen UNI + CONCH features (1536-dim), 50-gene panels, and the same training budget (100 epochs, patience 20, inner-validation selection). HCC, LYMPH_IDC, and READ excluded (see Appendix G).

Cohort	ST-Net NBE'20	Hist2ST BIB'22	HyperST CVPR'26	BLEEP NeurIPS'23	STEM ICLR'25	ContextST+CONCH
IDC Breast	0.551±0.01	0.309±0.07	0.567±0.01	0.519±0.00	0.439±0.01	**0.581±0.01**
COAD Colorectal	0.242±0.01	0.295±0.01	0.285±0.01	0.314±0.01	0.177±0.01	**0.318±0.01**
CCRCC Kidney	0.163±0.02	0.184±0.02	**0.256±0.02**	0.175±0.02	0.123±0.00	0.252±0.01
LUNG Lung	0.518±0.01	0.470±0.01	0.587±0.01	0.562±0.01	0.423±0.03	**0.589±0.01**
PAAD Pancreas	0.359±0.03	0.247±0.02	0.404±0.04	0.419±0.03	0.233±0.02	**0.437±0.03**
PRAD Prostate	0.294±0.02	0.333±0.02	0.384±0.02	0.348±0.01	0.207±0.02	**0.410±0.01**
SKCM Skin	0.665±0.01	0.522±0.03	0.653±0.01	0.645±0.01	0.491±0.04	**0.707±0.01**
Average	0.399	0.337	0.448	0.426	0.299	**0.470**

STEM: corrected DiT (T=1000, 34M params).

For the complete V3 model, the local output, global output, and broadcast slide representation are summed and added to the residual stream. A second pre-normalization residual update applies a two-layer feed-forward network with expansion factor four and GELU activation:

$$\widetilde{\mathbf{X}}^{(\ell)} = \mathbf{X}^{(\ell)} + \mathbf{L}^{(\ell)} + \mathbf{A}^{(\ell)} + \mathbf{G}^{(\ell)}, \tag{5}$$

$$\mathbf{X}^{(\ell+1)} = \widetilde{\mathbf{X}}^{(\ell)} + \mathrm{FFN}\Big(\mathrm{LN}(\widetilde{\mathbf{X}}^{(\ell)})\Big). \tag{6}$$

After L layers, layer normalization and a linear head map each spot representation to its G predicted gene values. The model produces the complete slide prediction in one forward pass.

3.4 Training objective

For a slide with measured expression $\mathbf{Y}$, we optimize mean squared error together with a gene-wise Pearson-correlation term:

$$\mathcal{L} = \mathrm{MSE}(\hat{\mathbf{Y}}, \mathbf{Y}) + \lambda \left(1 - \frac{1}{|\mathcal{V}|} \sum_{g \in \mathcal{V}} \rho_g \right), \tag{7}$$

where ρ_g is the Pearson correlation between predicted and measured expression across the N spots, $\mathcal{V}$ is the set of genes with nonzero target variance on that slide, and $\lambda = 0.5$. MSE constrains the scale of the predictions, while the correlation term encourages the model to reproduce spot-to-spot expression patterns.

The main model enables all three context pathways. Our factorial ablation independently toggles the local (L), global (G), and slide-level (S) pathways, avoiding conclusions tied to a particular order of architectural additions (Table 4).

4 Experiments

4.1 Datasets

We evaluate on HEST-1k Jaume et al. (2024). The controlled human-tissue benchmark retains seven cohorts representing seven organs; HCC, LYMPH_IDC, and READ are excluded for the data-quality and split constraints documented in Appendix G. Each cohort uses the provided patient-stratified folds. We predict 50 highly variable genes after $\log(1+x)$ normalization and use the same frozen UNI+CONCH features for every method in the main comparison.

Table 2: Cross-organ LOOO transfer, per-cohort Pearson ↑ (HEST-1k). Mean_{std} over three seeds; nested-validation. Leave-one-organ-out: trained on all other organ groups, tested on the held-out cohort. All methods use identical frozen UNI + CONCH features (1536-dim) and 50-gene panels, except STEM, for which a matched LOOO result is unavailable†. Best per row in bold.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	HyperST CVPR'26	BLEEP NeurIPS'23	STEM ICLR'25	ContextST+CONCH
IDC Breast	0.198±0.05	0.458±0.01	0.465±0.02	**0.529±0.02**	–†	0.520±0.02
COAD Colorectal	0.420±0.06	0.670±0.03	0.669±0.03	0.662±0.01	–†	**0.729±0.03**
CCRCC Kidney	0.068±0.01	0.055±0.01	0.106±0.01	0.095±0.01	–†	**0.109±0.01**
LUNG Lung	0.493±0.07	0.378±0.18	0.657±0.01	0.624±0.01	–†	**0.675±0.01**
PAAD Pancreas	0.459±0.02	0.399±0.02	0.463±0.00	0.537±0.02	–†	**0.548±0.01**
PRAD Prostate	0.079±0.00	0.098±0.02	0.114±0.03	0.118±0.01	–†	**0.118±0.01**
SKCM Skin	0.520±0.16	0.736±0.01	0.725±0.03	**0.748±0.01**	–†	0.728±0.04
Average	0.319	0.399	0.457	0.473	–†	**0.490**

†STEM was not evaluated in this feature-matched LOOO setting.

4.2 Experimental setup

Evaluation regimes. Intra-cohort evaluation trains a separate model within each cohort. Pooled five-fold cross-validation trains jointly across organs and holds out samples from represented organs. LOOO trains on six organs and tests the seventh, which is absent from training. Pooled folds are sample-disjoint; because patient identifiers are unavailable for some cohorts, we do not claim that they are patient-disjoint.

Metrics and protocol. We report the Pearson correlation coefficient (PCC), Spearman correlation (SCC), and mean squared error (MSE) between predicted and measured expression, averaged over folds and three random seeds (mean_{std}). Every model selects its checkpoint on an inner-validation split of the training fold and is evaluated once on the outer test-fold; where a cohort fold contains a single training slide, the inner split holds out a disjoint subset of that slide's spots. This removes the test-fold model selection present in earlier exploratory runs.

Baselines. We compare with ST-Net He et al. (2020), Hist2ST Zeng et al. (2022), HyperST Zhang et al. (2026), BLEEP Xie et al. (2023), and STEM Zhu et al. (2025) where results are available. Features, gene panels, splits, optimization budgets, and checkpoint selection are matched in the controlled tables. Published MCToGene results Sun et al. (2026), which use a different ten-cohort protocol, appear separately in Appendix F and are not treated as directly comparable.

Implementation. ContextST uses L=4 layers, hidden width d=256, 4 attention heads, k=8 neighbours, and 16 radial-basis functions (4.8M parameters). We train per slide with AdamW (learning rate 10^{-3}, weight decay 10^{-2}), a cosine schedule, and gradient clipping at 1.0, for at most 100 epochs with early-stopping patience of 20; each step samples at most 3,000 spots. Experiments run on one NVIDIA A100 GPU.

4.3 Intra-cohort prediction

Table 1 reports the feature-matched comparison. ContextST obtains the highest mean PCC (0.470), compared with 0.448 for HyperST and 0.426 for BLEEP, a 4.9% relative improvement over the strongest baseline. It leads on six of seven cohorts; HyperST is higher on CCRCC by 0.004. The gain is therefore broad but not uniform, and it is largest on skin and prostate.

4.4 Cross-organ generalization

LOOO is substantially harder because the target anatomy is absent from training. Nevertheless, ContextST reaches a mean PCC of 0.490 (Table 2), compared with 0.473 for BLEEP and 0.457 for HyperST. It leads on five of seven held-out cohorts; BLEEP is stronger on IDC and SKCM. The expanded comparison adds EGN, which is slightly higher on average

Table 3: Cross-organ POOLED, average Pearson ↑ (HEST-1k). 5-fold pooled cross-validation; mean_{std} over three seeds; nested-validation. Best in bold.

Method	HEST-1k
ST-Net He et al. (2020) (NBE'20)	$0.370_{0.01}$
Hist2ST Zeng et al. (2022) (BIB'22)	$0.737_{0.01}$
HyperST Zhang et al. (2026) (CVPR'26)	$0.756_{0.01}$
BLEEP Xie et al. (2023) (NeurIPS'23)	–
STEM Zhu et al. (2025) (ICLR'25)	–
ContextST (V3)	$\mathbf{0.760_{0.01}}$

at 0.500 (Table 11). Under pooled cross-validation, where organ types remain represented in training, ContextST reaches 0.760 (Table 3), narrowly above HyperST at 0.756. The gap between pooled and LOOO performance quantifies the remaining difficulty of unseen-organ transfer.

Appendix A reports a complementary seven-organ mouse experiment.

4.5 Context ablation and invariance

The factorial study in Table 4 evaluates all combinations of local (L), global (G), and slide-level (S) context on SKCM and LUNG. Local attention provides the largest gain: enabling it alone raises mean PCC from 0.608 to 0.640. Global and slide context have little effect without the local pathway, but the complete model achieves the highest mean (0.643). This result supports distance-biased neighbourhood aggregation as the primary contributor in the tested intra-cohort setting; broader claims about component importance under organ shift require a matched LOOO ablation.

Table 5 numerically verifies the analytic invariance result. Differences under translation and rotation remain below 6×10^{-5} and are attributable to floating-point evaluation; reflection is exact in this test.

Table 4: Factorial context ablation (intra-cohort, Pearson mean_{std} over three seeds on SKCM and LUNG). L, G, and S denote local attention, global attention, and slide representation. Local attention is the dominant contributor within an organ.

Variant	L	G	S	SKCM	LUNG	Avg
C000				$0.639_{0.00}$	$0.577_{0.01}$	0.608
C001			•	$0.633_{0.01}$	$0.578_{0.00}$	0.605
C010		•		$0.641_{0.01}$	$0.577_{0.00}$	0.609
C011		•	•	$0.639_{0.02}$	$0.583_{0.00}$	0.611
C100	•			$0.693_{0.01}$	$0.587_{0.00}$	0.640
C101	•		•	$0.695_{0.01}$	$0.585_{0.01}$	0.640
C110	•	•		$0.693_{0.01}$	$0.584_{0.00}$	0.638
C111	•	•	•	$\mathbf{0.696_{0.00}}$	$\mathbf{0.589_{0.01}}$	**0.643**

Table 5: Numerical coordinate-invariance check. Maximum absolute prediction difference when only the spot coordinates are transformed (randomly initialized model, 257-spot synthetic slide).

Transformation	Max. abs. difference
Translation	5.97×10^{-5}
Rotation	1.97×10^{-5}
Reflection	0
Spot permutation	5.96×10^{-7}

4.6 Spatial structure beyond gene-wise correlation

Gene-wise PCC does not test whether predictions preserve coherent tissue regions. We therefore cluster spots using either predicted or measured expression and compare the induced domains with the adjusted Rand index (ARI). On held-out slides, ContextST improves mean ARI from 0.148 for ST-Net to 0.170 (Appendix B). The large variation across slides cautions against interpreting this diagnostic as evidence of uniformly improved domain recovery.

4.7 Efficiency and scaling

ContextST contains 4.88M parameters and predicts a complete slide in one forward pass (Table 6). Its advantage is not minimum parameter count—ST-Net and BLEEP are smaller—but the combination of spatial context and direct inference. Model-only latency grows from 3.5 ms for 500 spots to 64.7 ms for 10,000 spots on an A100, with 3.1 GiB peak allocated memory at the largest size (Table 7). Feature extraction and disk I/O are excluded.

Table 6: Model complexity. Parameter counts and inference mechanisms.

Model	#Params (M)	Inference
ST-Net He et al. (2020) (NBE'20)	0.47	single-pass
Hist2ST Zeng et al. (2022) (BIB'22)	2.18	single-pass
HyperST Zhang et al. (2026) (CVPR'26)	3.99	single-pass
BLEEP Xie et al. (2023) (NeurIPS'23)	0.54	single-pass
STEM Zhu et al. (2025) (ICLR'25)	34.16	iterative
ContextST (V3)	4.88	single-pass

Table 7: ContextST inference scaling on an NVIDIA A100 80GB. Median model-only latency and peak allocated GPU memory over synthetic slides; feature extraction and disk I/O are excluded.

Spots	Latency (ms)	Memory (MiB)
500	3.48	43
1,000	3.63	67
3,000	9.36	335
5,000	21.31	842
10,000	64.74	3,183

5 Discussion

The results separate two conclusions that are often conflated. First, a compact direct predictor can be competitive when morphology features, gene panels, splits, and model selection are controlled. Second, pooled multi-organ performance is not a substitute for unseen-organ evaluation: the marked drop from pooled to LOOO PCC shows that anatomical shift remains difficult. The ablation further suggests that explicit local geometry is more useful than adding unconstrained global context alone in the two tested intra-cohort datasets.

Direct and generative models serve different purposes. STEM Zhu et al. (2025) models a conditional expression distribution and can represent one-to-many uncertainty through iterative sampling. ContextST instead produces a point estimate optimized for correlation and squared error. Its benefits are deterministic, single-pass inference and explicit spatial invariance; they should not be interpreted as evidence that expression is uniquely determined by morphology.

Limitations. The experiments predict 50 genes from frozen UNI+CONCH embeddings, so they do not establish performance for transcriptome-scale output or end-to-end image encoding. LOOO still uses a gene panel defined by the benchmark and does not cover shifts in platform, institution, or species simultaneously. Coordinate invariance holds only for the spatial pathway with fixed image features; rotating the histology image can change its frozen embedding. The domain-recovery analysis covers only ST-Net and has high variance. Finally, pooled folds are sample-disjoint but cannot be verified as patient-disjoint where patient identifiers are unavailable.

6 Conclusion

We presented ContextST, a local–global transformer for predicting spatial gene expression from histology. Pairwise-distance encoding gives its spatial pathway invariance to rigid coordinate transformations, while local, global, and slide-level context support complete-slide prediction in a single pass. Under matched HEST-1k evaluation, ContextST improves mean intra-cohort Pearson correlation and remains competitive in LOOO transfer, although the pooled-to-LOOO gap highlights the difficulty of anatomical transfer. These results establish ContextST as an efficient point-prediction baseline and motivate future work that combines its geometric structure with calibrated generative uncertainty.

AI use statement

Generative AI tools assisted with manuscript organization, language editing, ICLR formatting, and consistency checks between narrative claims and reported tables. They were not used to generate experimental data or execute model experiments. The authors reviewed all AI-assisted revisions and take responsibility for the final content of this work.

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A Mouse cross-organ diagnostic

We evaluate 59 mouse Visium slides spanning seven organs with mouse-specific 50-gene panels. Pooled performance reaches 0.887, whereas every method remains near the floor under LOOO; we therefore do not interpret the latter as evidence of successful cross-species organ transfer.

Table 8: Cross-organ generalization on mouse HEST (three seeds, mean_{std}). looo: leave-one-organ-out transfer; pooled: five-fold pooled cross-validation. PCC↑ on 50 mouse HVGs.

Method	looo PCC†	pooled PCC
ST-Net He et al. (2020)	$0.013_{0.011}$	$0.357_{0.036}$
Hist2ST Zeng et al. (2022)	$0.018_{0.007}$	$0.829_{0.028}$
HyperST Zhang et al. (2026)	$\mathbf{0.019_{0.006}}$	$0.865_{0.013}$
BLEEP Xie et al. (2023)	$-0.006_{0.003}$	$0.848_{0.012}$
STEM Zhu et al. (2025)	–	$0.279_{0.009}$
ContextST	$0.018_{0.006}$	$\mathbf{0.887_{0.005}}$

† All looo values are near the floor because the organ-specific gene panels do not transfer reliably.

B Spatial-domain recovery

Table 9: Spatial-domain recovery. ARI between domains obtained from predicted and measured expression on held-out slides (mean_{std} over three seeds; higher is better).

Method	ARI ↑
ST-Net He et al. (2020)	$0.148_{0.13}$
ContextST	$\mathbf{0.170_{0.15}}$

C Full per-cohort results

Table 10 gives the per-cohort breakdown for all seven retained cohorts (HCC/liver, LYMPH_IDC, and READ excluded; see Appendix G) across all six baselines and ContextST.

D Cross-organ LOOO transfer, expanded baselines (UNI + CONCH)

Table 11 extends Table 2 with M2OST and EGN under the same UNI+CONCH protocol. EGN attains the highest mean PCC (0.500), followed by ContextST at 0.490; their per-cohort ranking varies.

E Cross-organ LOOO transfer with UNI features

Table 12 reports a complementary LOOO comparison using UNI features. EGN has the highest mean PCC (0.493), while ContextST reaches 0.477. M2OST narrowly leads EGN on LUNG (0.667 versus 0.666). STEM uses the corrected DiT configuration with $T = 1000$; its SKCM entry contains one completed seed and is marked accordingly.

Table 10: Full per-cohort intra-cohort results on HEST-1k (Pearson mean_{std} over three seeds; nested-validation). All methods use identical frozen UNI + CONCH features (1536-dim) and 50-gene panels, except ContextST (V3) which uses UNI (1024-dim) for comparison. HCC (liver), LYMPH_IDC, and READ are excluded per Appendix G. Best per row in bold.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	HyperST CVPR'26	BLEEP NeurIPS'23	M2OST AAAI'25	EGN WACV'23	STEM ICLR'25	ContextST (V3)	ContextST+CONCH
IDC Breast	0.551±0.01	0.309±0.07	0.567±0.01	0.519±0.00	0.571±0.01	0.551±0.01	0.439±0.01	0.567±0.00	**0.581±0.01**
COAD Colorectal	0.242±0.01	0.295±0.01	0.285±0.01	0.314±0.01	**0.328±0.01**	0.293±0.01	0.177±0.01	0.299±0.00	0.318±0.01
CCRCC Kidney	0.163±0.02	0.184±0.02	0.256±0.02	0.175±0.02	0.210±0.02	0.223±0.02	0.123±0.00	**0.265±0.02**	0.252±0.01
LUNG Lung	0.518±0.01	0.470±0.01	0.587±0.01	0.562±0.01	**0.598±0.00**	0.584±0.01	0.423±0.03	0.589±0.01	0.589±0.01
PAAD Pancreas	0.359±0.03	0.247±0.02	0.404±0.04	0.419±0.03	0.413±0.04	0.415±0.03	0.233±0.02	0.392±0.04	**0.437±0.03**
PRAD Prostate	0.294±0.02	0.333±0.02	0.384±0.02	0.348±0.01	**0.413±0.01**	0.377±0.03	0.207±0.02	0.390±0.02	0.410±0.01
SKCM Skin	0.665±0.01	0.522±0.03	0.653±0.01	0.645±0.01	0.654±0.01	0.631±0.01	0.491±0.04	0.696±0.00	**0.707±0.01**
Average	0.399	0.337	0.448	0.426	0.455	0.439	0.299	0.457	**0.470**

ContextST (V3) uses UNI (1024-dim); all other methods use UNI + CONCH (1536-dim). STEM: corrected DiT (T=1000, 34M params).

Table 11: Cross-organ LOOO transfer, expanded baselines, per-cohort Pearson ↑ (HEST-1k) (Pearson mean_{std} over three seeds; nested-validation; liver excluded). Leave-one-organ-out: trained on all other organ groups, tested on the held-out cohort. All methods use identical frozen UNI + CONCH features (1536-dim) and 50-gene panels, except STEM, for which a matched result is unavailable†. Best per row in bold.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	HyperST CVPR'26	BLEEP NeurIPS'23	M2OST AAAI'25	EGN WACV'23	STEM ICLR'25	ContextST+CONCH
IDC Breast	0.198±0.05	0.458±0.01	0.465±0.02	0.529±0.02	0.437±0.04	**0.567±0.02**	–†	0.520±0.02
COAD Colorectal	0.420±0.06	0.670±0.03	0.669±0.03	0.662±0.01	0.657±0.05	0.728±0.01	–†	**0.729±0.03**
CCRCC Kidney	0.068±0.01	0.055±0.01	0.106±0.01	0.095±0.01	0.081±0.03	0.103±0.02	–†	**0.109±0.01**
LUNG Lung	0.493±0.07	0.378±0.18	0.657±0.01	0.624±0.01	0.665±0.01	0.672±0.01	–†	**0.675±0.01**
PAAD Pancreas	0.459±0.02	0.399±0.02	0.463±0.00	0.537±0.02	0.513±0.03	**0.572±0.01**	–†	0.548±0.01
PRAD Prostate	0.079±0.00	0.098±0.02	0.114±0.03	0.118±0.01	0.087±0.02	0.107±0.01	–†	**0.118±0.01**
SKCM Skin	0.520±0.16	0.736±0.01	0.725±0.03	0.748±0.01	0.740±0.01	**0.750±0.02**	–†	0.728±0.04
Average	0.319	0.399	0.457	0.473	0.454	**0.500**	–†	0.490

†STEM was not evaluated in this feature-matched LOOO setting.

F Published Per-Cohort Comparison

G HEST-1k Cohort Statistics

Table 14 lists the number of samples, total spots, and cross-validation folds for each HEST-1k cohort. Two organ groups in HEST-1k contain multiple cohorts: breast (IDC + LYMPH_IDC) and colorectal (COAD + READ). Within each group the constituent cohorts differ substantially in size: IDC has 2.25× more spots than LYMPH_IDC (44,974 vs. 19,964), and COAD has 2.2× more spots than READ (18,523 vs. 8,407). To avoid misrepresentation from unweighted averaging over unequal cohorts, Table 1 uses IDC as the representative breast cohort and COAD as the representative colorectal cohort, discarding the smaller partner cohorts (LYMPH_IDC and READ) from the organ-level rows. This choice maximises statistical reliability of each organ-level estimate.

Table 12: Cross-organ LOOO transfer with UNI features (Pearson mean_{std} over three seeds; nested-validation; HCC, LYMPH_IDC, and READ excluded). Leave-one-organ-out: trained on all other organs, tested on the held-out organ. Best per row in bold.

Organ	ST-Net NBE'20	Hist2ST BIB'22	HyperST CVPR'26	BLEEP NeurIPS'23	M2OST AAAI'25	EGN WACV'23	STEM ICLR'25	ContextST (V3)
IDC Breast	0.217±0.06	0.459±0.02	0.501±0.02	0.509±0.03	0.452±0.01	0.531±0.03	0.406±0.01	**0.534±0.02**
COAD Colorectal	0.430±0.03	0.688±0.04	0.669±0.06	0.667±0.02	0.672±0.03	**0.749±0.01**	0.504±0.03	0.703±0.01
CCRCC Kidney	0.084±0.03	0.084±0.01	0.108±0.01	0.100±0.02	0.122±0.01	0.106±0.01	0.090±0.01	**0.124±0.00**
LUNG Lung	0.504±0.04	0.499±0.06	0.650±0.01	0.621±0.02	**0.667±0.01**	0.666±0.01	0.561±0.01	0.660±0.01
PAAD Pancreas	0.404±0.05	0.360±0.07	0.397±0.03	0.521±0.02	0.522±0.01	**0.551±0.02**	0.460±0.02	0.507±0.04
PRAD Prostate	0.071±0.01	0.104±0.02	0.092±0.03	0.111±0.00	0.107±0.02	0.098±0.02	0.068±0.01	**0.119±0.02**
SKCM Skin	0.402±0.08	0.652±0.08	0.720±0.02	0.745±0.01	0.724±0.02	**0.749±0.01**	0.686†	0.692±0.06
Average	0.302	0.407	0.448	0.468	0.466	**0.493**	0.396†	0.477

†STEM DiT (corrected, T=1000, 100 epochs) LOOO: SKCM (skin) has only 1 seed complete; all other organs use 3 seeds. Average marked † accordingly.

Table 13: Published per-cohort accuracy on the standard HEST-1k splits, mean PCC↑ as reported in respective papers (ten cohorts, no organ grouping). Shown for context only—not comparable to Table 1, which uses organ grouping and nested-validation. †ContextST uses our organ-grouped splits, not the standard per-cohort split.

Method	HEST-1k (10 cohorts)
ST-Net He et al. (2020) (NBE'20)	0.286
Hist2ST Zeng et al. (2022) (BIB'22)	–
HyperST Zhang et al. (2026) (CVPR'26)	–
MCToGene Sun et al. (2026) (CVPR'26)	**0.435**
ContextST† (ours)	0.440

Table 14: HEST-1k cohort statistics. Spots counted after barcode filtering. Folds = number of train/test splits in the per-cohort cross-validation. HCC (liver) is excluded from Table 1 due to its near-zero signal floor.

Cohort	Organ	Samples	Spots	Folds
IDC	Breast	4	44,974	4
LYMPH_IDC	Breast	4	19,964	4
COAD	Colorectal	4	18,523	2
READ	Colorectal	4	8,407	2
CCRCC	Kidney	24	74,220	6
HCC	Liver*	2	4,248	2
LUNG	Lung	2	7,505	2
PAAD	Pancreas	3	9,793	3
PRAD	Prostate	23	62,710	2
SKCM	Skin	2	5,716	2

*HCC (liver) is included here for completeness but excluded from Table 1; all baselines score near zero on this cohort due to low spatial gene-expression variance.